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Preferential Pathways Inversion From Cross-Borehole Electrical Data

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Key Points:

- Standard modeling strategies are not well suited to identify narrow geological structures from cross-borehole electrical data inversion
- Inversion scheme based on discrete-dual-porosity forward simulations enable to image narrow preferential flow-paths from electrical data
- Prior assumptions are needed to invert the position and length of narrow preferential flow-paths

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Abstract Identification of preferential flow-paths, such as fractures, is required for various issues in geosciences. When chemicals are injected into the subsurface, monitoring the resulting structural and chemical changes remains a challenge. The ability of geophysical tomography to tackle this problem is not fully explored due to the lack of numerical methods suitable for modeling narrow structures. We explore how discrete representation of preferential flow-paths provides innovative ways to invert electrical resistivity data collected during reagent injection at a contaminated site. The data set is inverted with a scheme where a new fracture is added at every iteration. This allows identifying newly created narrow conductive structures from the field data collected before and after injection. Fracture location remains overall consistent despite using different starting points for the fracture search. A prior constraint on fracture length improves convergence. These results show the potential of discrete inversion for identifying narrow structures from electrical resistivity monitoring.

Plain Language Summary In some geoscience activities, such as water extraction, geological storage, and contaminant fate, chemical solutions are injected into wells, which potentially leads to modify the system properties. Identifying preferential flow-paths is therefore necessary even though monitoring structural and chemical changes in the subsurface remains a challenge. Electrical geophysical imaging techniques may be adapted to provide information for this purpose. However, their potential is not fully explored due to the limitations of numerical models to represent small-scale structures. In this work, we explore the representation of preferential flow-paths by discrete elements for the inversion of electrical resistivity data collected during a reagent injection at a contaminated site. The inversion scheme based on discrete forward simulations successively identifies small-scale structures and focuses on evaluating their localization and length. The results show the need for further efforts to use discrete electrical resistivity inversion in the presence of narrow structures.

1. Introduction

Fractures, or thin localized flow-paths, greatly influence subsurface transport properties and are determinant structures for many challenges in geosciences (e.g., Berkowitz, 2002; Neuman, 2005). As they control the transport of fluids, their presence is an advantage for water extraction, while it increases the risk of leakage in the context of geological storage and enhances the propagation of contaminant. These structures are naturally present in the subsurface, but might also be created or reopened (accidently or intentionally) during the injection of fluids in boreholes (e.g., Singhal & Gupta, 2010). Regardless of how they are created, identifying fractures in the underground, and potentially at the borehole scale, is required for predicting flow and transport behavior in the subsurface, establishing risk assessment strategies, and developing appropriate management and decision making plans (e.g., Gallegos et al., 2015; Harvey et al., 2013; Vidic et al., 2013).

Hydrogeological information are typically acquired from borehole observations, including piezometer measurements, pumping tests, and flowmeter experiments, which inform on the properties of fractures that intersect one or several boreholes (e.g., Fischer et al., 2018; Le Goc et al., 2010; Roubinet et al., 2015). Geophysical methods, such as temperature, self-potential, and electrical resistivity tomography, are also used to characterize these structures from borehole and surface measurements (e.g., DesRoches et al., 2017; Klepikova et al., 2022; Robinson et al., 2016). Amongst these methods, electrical resistivity tomography (ERT) presents the advantage of being well suited to monitor the injection of electrically conductive fluids in heterogeneous environments (e.g., Leslie & Heinse, 2013; Lévy, Thalund-Hansen, Bording, Fiandaca, Christiansen, et al., 2022; Robert et al., 2011; L. D. Slater et al., 1997), while providing a large coverage of the considered systems and being usually less

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expensive and intrusive than hydrogeological methods. ERT experiments provide thousands of data that must be systematically inverted with efficient forward modeling approaches incorporated into well-suited inversion scheme (e.g., Johnson et al., 2022; Wagner & Uhlemann, 2021). Despite the wide range of methods available to invert these data, the detection of narrow geometries is limited by the measurement scale and inversion smoothing constraints that homogenize small-scale features (e.g., Auken et al., 2015; Parsekian et al., 2015; Rücker, 2011). Although a few methods are able to detect the presence of flow-paths from ERT data, they present limitations related to the detection of accurate shapes. Some of them explicitly integrate the (assumed to-be-known) shape and position of the structure of interest (fracture or barrier) in the inversion scheme (e.g., Robert et al., 2012; Robinson et al., 2013; L. Slater & Binley, 2006), others rely on the accurate selection of specific inversion parameters for time-lapse data (e.g., Nguyen et al., 2016).

On the other hand, both field experiments (e.g., Boadu et al., 2005; Busby, 2000; Lane et al., 1995) and forward numerical simulations (e.g., Berryman & Hoversten, 2013; Beskardes & Weiss, 2018; Demirel et al., 2018) show that the presence of fractures results in significant anomalies in the apparent electrical resistivity for various electrode configurations and system properties. The latter numerical results are obtained with forward modeling approaches that simulate electrical current propagation in fractured rocks and rely on various assumptions related to effective-medium theory and fracture network representation. Amongst these methods, the 2.5D discrete-dual-porosity (DDP) approach developed by Caballero-Sanz et al. (2017) combines (a) a discrete representation of thin conductive structures, inspired by the discrete fracture network (DFN) representation used in fractured-rock applications, (b) a dual-porosity formulation of the exchanges between fractures and matrix, and (c) a 2.5D formulation of the problem to accurately and efficiently simulate current flow between point electrodes. The DDP approach results in an explicit representation of the fractures with optimized computational cost. To our knowledge, the present work is the first attempt to invert electrical data in terms of explicitly represented narrow localized heterogeneities.

In this context, our work focuses on evaluating the ability of the 2.5D DDP approach to be used in inversion scheme for identifying the presence of narrow conductive structures from cross-borehole ERT (XB-ERT) data. To this end, we consider field data collected before and during the remediation of a contaminated site in Denmark (Lévy, Thalund-Hansen, Bording, Fiandaca, Christiansen, et al., 2022) and we intend to localize thin conductive structures revealed by the injection. While standard inversion strategies rely on estimating the properties of a large number of discretization cells, inverting ERT data with the 2.5D DDP approach requires to estimate the properties of a few discrete structures. This greatly reduces the number of parameters that must be estimated. Using a sequential inversion scheme where a new fracture is added at each iteration, thin conductive structures, whose presence reduces the misfit between observed and simulated data, are progressively added to our system representation. For this first step in inverting cross-borehole ERT data with a discrete inversion scheme, the electrical conductivity of the surrounding matrix and the narrow flow-paths are considered homogeneous and estimated from the data acquired before injection and direct observations, respectively. The inversion method focuses on the location and length of the conductive narrow structures identified after the injection experiment.

After describing the considered site, data, and numerical methods in Section 2, the results are presented in Section 3, and conclusions are provided in Section 4.

2. Materials and Methods

2.1. Site and Experiment Description

The study site is located in Farum (Denmark) and is part of a pilot project where a combination of zero-valent iron (ZVI), carbon substrate, and bacteria was injected as remediation agent (reagent) to test the in situ degradation of perchlorethylene contamination. The remediation target was a volume comprised between 10 and 20 m depth, where the geology consists of sandy till at 10–13 m and meltwater sand below. As ZVI and associated dissolution products are characterized by low electrical resistivity compared to the host medium (L. D. Slater et al., 2005), time-lapse XB-ERT surveys were performed to monitor the injection. Data were acquired along three 2D transects in nine boreholes, each containing 32 electrodes with vertical spacing of 0.3 m (Lévy, Thalund-Hansen, Bording, Fiandaca, Christiansen, et al., 2022). Single and cross-borehole configurations were used, the latter including equatorial configurations where current injection and voltage measurements are performed in-between boreholes, and parallel configurations where current injection is performed in one borehole and voltage measurements in another borehole (T. Bording et al., 2021; Lévy, Thalund-Hansen, Bording, Fiandaca, Christiansen,

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et al., 2022; Zhou & Greenhalgh, 2001). This data set was collected before injection and regularly afterward over 2 years (Lévy, Thalund-Hansen, Bording, Fiandaca, Christiansen, et al., 2022). Here, we only consider the baseline (before injection) and the data set 1 day after injection. They are also completed by the analysis of post-injection core samples that shows the presence of 3 cm-thick iron-rich sand horizons related to the injection (Lévy, Thalund-Hansen, Bording, Fiandaca, Christiansen, et al., 2022).

2.2. Standard Data Inversion

In the present study, we focus on one 2D transect with 4 boreholes located 2.5–3 m apart along a plane. A total of 2,618 datapoints were collected after the injection, and 2,232 before. Each datapoint measures the difference in electrical potential between two voltage electrodes while injecting electrical current between two current electrodes. After removing outliers as described in Lévy, Thalund-Hansen, Bording, Fiandaca, Christiansen, et al. (2022), 2008 datapoints remain for both data sets.

These data sets are inverted with AarhusInv software (Auken et al., 2015) using the borehole electrode implementation presented in T. S. Bording et al. (2019). A regular model grid composed of 0.15 m × 0.5 m cells with 99 and 29 cells in the vertical and horizontal directions respectively is considered, corresponding to the inversion of 2,871 parameters (Lévy, Thalund-Hansen, Bording, Fiandaca, Christiansen, et al., 2022). Roughness constraints imply that the resistivity of neighboring model cells is allowed to vary within a factor of 1.1 and 1.075 in the vertical and horizontal direction, respectively, as in T. Bording et al. (2021). Note that vertical constraints are typically tighter for inversion of cross-borehole resistivity data, compared to surface data, to avoid artifacts. In addition, a time-lapse inversion scheme constrains the model 1 day after injection to the baseline model through the asymmetric minimum support norm, as described by Fiandaca et al. (2015).

2.3. Discrete Inversion Scheme

The DDP model relies on the working hypothesis that small-scale heterogeneities are present on the studied site. Using discrete representation of narrow conductive structures in inversion strategies is performed by incorporating the 2.5D DDP forward modeling approach developed by Caballero-Sanz et al. (2017) in an inversion framework. The procedure focuses on the presence of horizontal fractures, most likely created during the high-pressure injection experiment (Lévy, Thalund-Hansen, Bording, Fiandaca, Christiansen, et al., 2022). Each fracture is defined by the coordinates of its center (x, y), length l, aperture b and electrical conductivity σ_f. The parameters (x, y) and l are inverted for each fracture, while b and σ_f are set to constant values (cf. Section 2.6). The iterative procedure considers the addition of a new fracture to the system at each increment. The procedure stops after 10 successive iterations resulting in rejection of the new fracture. The inversion scheme is therefore irreversible as a fracture can only be added (not removed). The iterative loop on the number of potential fracture encloses an inversion loop that defines the set of parameters (x, y, l) minimizing the misfit Θ (Section 2.4). Due to the complexity of the misfit function, standard gradient-based methods fail to perform this minimization, leading to the use of a direct search optimization method (e.g., Audet & Dennis, 2002; Kolda et al., 2003; Lewis et al., 2007). Thus, a succession of inversions (fracture by fracture) is run and converges accordingly to the internal stopping criteria of the direct search algorithm (Audet & Dennis, 2002; Kolda et al., 2003; Lewis et al., 2007). The search starts with random values of the initial set of parameters (x_0, y_0, l_0) following an uniform distribution. If the misfit obtained at the end of the direct search optimization Θ^i_{final} of iteration i is smaller than the final misfit Θ^{i-1}_{final} of the previous iteration, the new fracture is accepted and incorporated into the DDP model.

2.4. Misfit Definition for Discrete Inversion

The inversion methods presented in Section 2.3 rely on the following definition of the data misfit Θ:

$$\Theta = \frac{1}{N}\sum_{j=1}^{N}\left(\Delta^j_{sim} - \Delta^j_{obs}\right)^2, \tag{1}$$

where N is the number of datapoints inverted. Δ^j_{sim} and Δ^j_{obs} correspond to the apparent resistivity simulated and observed, respectively, for quadrupole j. In the discrete inversion method, our objective is to identify the presence of fractures due to the injection. We consider two distinct models that are the numerical counterparts of the

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Table 1
Fixed and Inverted Parameters, for Length1 and Length2 Cases, Used in the Discrete Inversion Scheme

	Parameter notation	Value	Limitations
Fixed parameters	Matrix conductivity, σ_m	0.08 S/m	Homogeneous
	Fracture conductivity, σ_f	2 S/m	Same for all fractures
	Fracture aperture, b	0.03 m	
	Fracture dip, α	0°	
Inverted parameters	Fracture length, l	*Length1*	Critical influence of the first fractures (irreversibility)
	Fracture position, (x, y)		
	0 m $\leqslant l \leqslant$ 2 m	*Length2*	
	Fracture position, (x, y)		

Note. The last column provides the limitations associated with the considered parameters. "*Length1*" and "*Length2*" are the names of the two length settings tested in this work. "*Length1*" has an unconstrained length, while "*Length2*" has a constrained length between 0 and 2 m. σ_m: electrical conductivity of the matrix (in S/m). σ_f: electrical conductivity of the fracture, that is, of the fluid flowing through the fracture (in S/m). b: aperture of the fracture (in m). α: dip angle of the fracture (in degree). l: length of the fracture (in m). (x, y): x coordinate of the center of the fracture, y coordinate (depth) of the center of the fracture (both in m).

subsurface system before and after injection and correspond to two porous models without and with fractures. For each datapoint j, ρ^j_{porous} and ρ^j_{fract} are the apparent resistivities of the synthetic porous domains without and with fracture, respectively, and ρ^j_{init} and ρ^j_{inj} are the in situ counterparts, corresponding to the apparent resistivities observed before and after the injection, respectively. The anomalies due to the presence of fractures after injection are expressed with the following normalized expressions of Δ^j_{sim} and Δ^j_{obs}:

$$\Delta^j_{sim} = \frac{\rho^j_{fract} - \rho^j_{porous}}{\rho^j_{porous}}, \qquad \Delta^j_{obs} = \frac{\rho^j_{inj} - \rho^j_{init}}{\rho^j_{init}}, \tag{2}$$

that are used in the misfit Definition 1. In other words, the discrete inversion method presented in Section 2.3 relies on a misfit definition that focuses on anomalies in apparent resistivity due to the presence of fractures after injection. Note that working with differences and normalized values enable one to highlight changes between the initial and modified properties of the considered system, as done in existing studies (e.g., Auken et al., 2014; Daily & Owen, 1991; Doetsch et al., 2012; LaBrecque & Yang, 2001). Here, it results in removing the effect of the surrounding matrix heterogeneity in order to focus on the influence of fractures.

2.5. Data Processing

For simplicity, we have chosen to rely on the filtering describes in Lévy, Thalund-Hansen, Bording, Fiandaca, Christiansen, et al. (2022), even though datapoints removed by the processing for AarhusInv inversion could have been kept for this inversion. Then, datapoints for which the quadrupole is not common to the baseline and post-inversion measurements are removed, so that the same quadrupoles are involved in the ratio calculations. A total of 1,883 values remains. A two-step processing of the simulated data is performed at each iteration of the discrete inversion scheme. First, negative values of the simulated ρ^j_{porous} and ρ^j_{fract} are removed. Although negative resistivity values have been related to the entrance (exit) of the electrical current in (out) the fracture (e.g., Demirel et al., 2018), they are removed here to be consistent with the processing done in Lévy, Thalund-Hansen, Bording, Fiandaca, Christiansen, et al. (2022). Second, because we focus on the creation of new fractures after the injection (Section 2.4), only negative values of Δ^i_{sim} are considered based on the assumption that no increase in resistivity is expected following the injection, as done in Johnson et al. (2022).

2.6. Parameter Settings

The model parameters used in the discrete inversion scheme are summarized in Table 1. The DDP-based model could handle the inversion of all the parameters. However, to save computation time, hypothesis are needed to

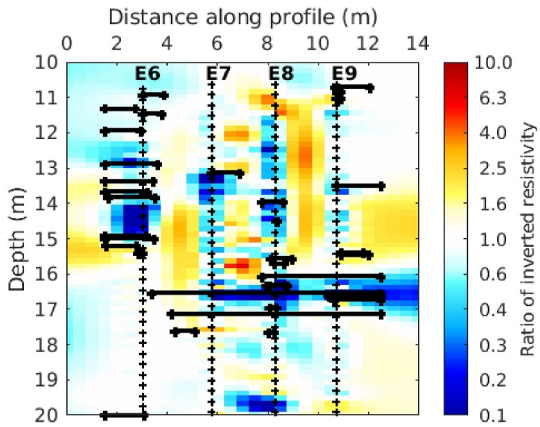


Figure 1. Background map (blue to red pixels): ratio of apparent resistivity data (post-injection/baseline) inverted by AarhusInv (Figure S9.1, Supporting information to Lévy, Thalund-Hansen, Bording, Fiandaca, Christiansen, et al. (2022)). Black lines: fractures inverted by the discrete inversion approach with *Length1* setting (Table 1). Black crosses correspond to the electrodes in the four boreholes E6, E7, E8, and E9.

simplify the model, which goes with potential limitations (last column of Table 1, details in Section 3). The electrical conductivity of the homogeneous matrix σ_m is obtained by inverting the data before injection with a homogeneous porous domain. The fracture electrical conductivity σ_f is inferred from Figure 8 in Lévy, Thalund-Hansen, Bording, Fiandaca, Christiansen, et al. (2022), where pixel conductivity is converted into fracture conductivity by using the ratio of pixel height to fracture aperture and of pixel width to core diameter, resulting in a conductivity range from 1 to 3.3 S/m. The fracture aperture b is determined from the iron-rich sand horizons observed in post-injection geological cores (Section 2.1). The fracture dip is assumed to be horizontal considering the results for post-injection data obtained with AarhusInv (Figure 2 in Lévy, Thalund-Hansen, Bording, Fiandaca, Christiansen, et al. (2022)). Inverting the position and length of fractures is performed for two length settings, unconstrained (*Length1*) and constrained (*Length2*). The configurations are defined to test the impact of the irreversibility of the discrete inversion scheme as a poorly constrained fracture length may introduce an irreversible bias. The choice of the constrained length range (between 0 and 2 m) is related to the injection borehole distance and to the results obtained with the unconstrained conditions (Section 3.1). *Length1* converges after 70 iterations resulting in adding 36 fractures to the distribution. *Length2* converges after 85 iterations resulting in adding 54 fractures to the distribution. They lead to a maximum of 210 and 255 inverted parameters respectively.

3. Results

3.1. Comparison of Inversion Methods

Former results obtained with AarhusInv software (Lévy, Thalund-Hansen, Bording, Fiandaca, Christiansen, et al., 2022) are shown at the background of Figure 1. Pixels with a low resistivity ratio (in blue), that is, higher conductivity after injection, indicate the location of the reagent. They form localized conductive layers with a thickness varying from centimeter to meter. DDP-inverted fractures, shown as black lines in Figure 1, are mainly located along the E6 borehole with an average length of 2 m, and at 16 m depth with a length exceeding 4 m across boreholes E7, E8, and E9. A few additional fractures are spread over the whole domain. DDP-inversion cannot be directly compared with standard inversion as the number of parameters differs by an order of magnitude (210 parameters inverted using the discrete scheme vs. 2,871 parameters with AarhusInv software). Nevertheless, DDP-inverted fractures are mainly overlapping the conductive layers of the standard inversion. These results, obtained with the unconstrained length setting, show that DDP-fractures inversion is consistent with the interpretation of the standard inversion and the post-injection geological cores (Lévy, Thalund-Hansen, Bording, Fiandaca, Christiansen, et al., 2022), and they numerically confirm that the XB-ERT data set is compatible with the presence of fractures. They also suggest the presence of long DDP-inverted fractures having a length up to 7 m that are analyzed in Section 3.2.

3.2. Analysis of the Discrete Inversion Scheme

The results of the discrete inversion scheme are first analyzed by comparing the fracture distribution (Figure 2a) and final misfit evolution (Figure 2b) for the unconstrained (*Length1*) and constrained (*Length2*) length settings. With *Length1* setting, out of 70 iterations, 36 fractures are accepted and 34 are rejected (Figure 2b). With *Length2* setting, out of 85 iterations, 54 fractures are accepted and 31 are rejected. The larger number of rejected fractures and the non regular decrease of Θ_{final} observed for *Length1* setting are induced by (a) increasing the parameter space (compared to *Length2*), (b) the irreversibility of the inversion scheme, and (c) the numerous local minima of the misfit function (Section 2.4). Even though similar spatial distribution of fractures are observed in both cases, the presence of long fractures with *Length1* is not reproduced with *Length2* (Figure 2a). This suggests that the presence of long fractures for unconstrained length is due to the irreversibility of the scheme where non optimal fractures are kept in the following iterations. The latter limits the misfit reduction compared to shorter fractures. Consequently, the unconstrained case (*Length1*) presents the following drawbacks: (a) the direct search

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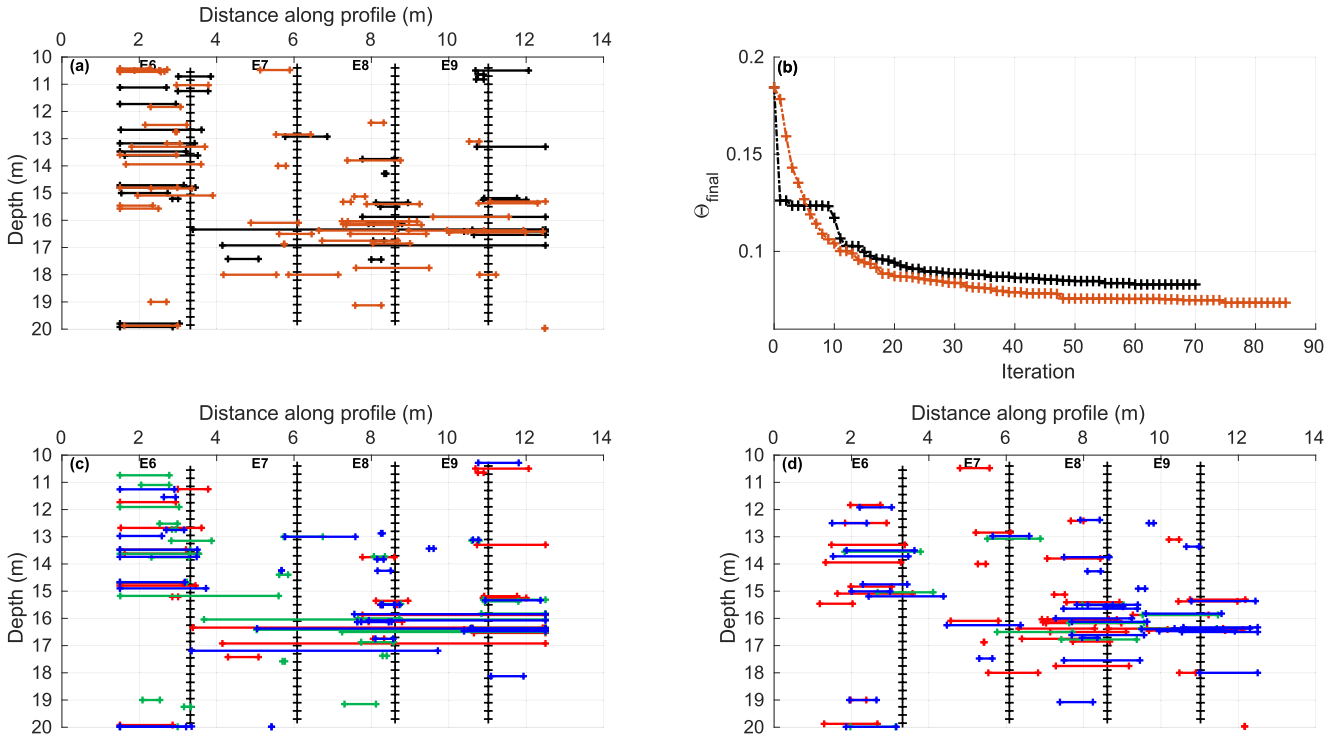


Figure 2. Comparison of *Length1* setting (unconstrained length, in black) with *Length2* (constrained length, in orange). (a) Fracture distributions for both settings, for one starting point. (b) Comparison of the final misfits. Panels (c and d) comparison of three distributions when varying the starting point sequence (respectively in red, green, and blue) for *Length1* (c) and *Length2* (d). In (a), (c), and (d) black crosses correspond to the electrodes and E6, E7, E8, and E9 refer to the measurement boreholes.

optimization does not converge at each iteration to a global minimum mainly because of numerous local minima and (b) the inversion scheme gives a large weight to the first fractures added to the model, in an irreversible manner, which greatly impacts the misfit optimization at the next iterations.

The irreversibility of the inversion scheme is further analyzed by testing the impact of the starting points on the final fracture distribution. The fracture distributions for both unconstrained and constrained fracture length, and three different starting point sequences, are shown in Figures 2c and 2d. Similar distributions are observed for all models. In particular, two zones show a high density of fractures: along E6 between 10.5 and 15.5 m depth, and across E7 and E9 between 15.8 and 17 m depth. The comparison between *Length1* (Figure 2c) and *Length2* (Figure 2d) shows that the inversion with the unconstrained length setting is more sensitive to the starting point. For example, the depth of long fractures varies between 16 and 17 m depth depending on the starting point. Thus, (a) the overall fracture distribution does not vary significantly between the tests (two dense areas are found), (b) fractures in the densest areas are more likely to exist, and (c) the DDP inversion is sensitive to starting points, particularly when the length is unconstrained, because of its iterative architecture and the local investigation of the direct search optimization.

3.3. Overall Data Fitting

In this section, the apparent resistivity simulated by the DDP-fracture inversion is compared to the observed apparent resistivity collected on field. Post-injection resistivity data are generally lower than baseline data (Figure 3a) due to the injection of the highly conductive reagent. To assess the suitability of fracture modeling, we intend to artificially reproduce the observed post-injection data by adding the effects of simulated fractures to the observed baseline (Figure 3b). It is an indirect way to compare observations to simulations. A total of 55% of the simulated data fit the observed post-injection data within 2 Ωm. The prediction improvement observed in Figure 3b compared to Figure 3a represents the part of the data which can be interpreted by the presence of preferential thin pathways filled with the conductive reagent.

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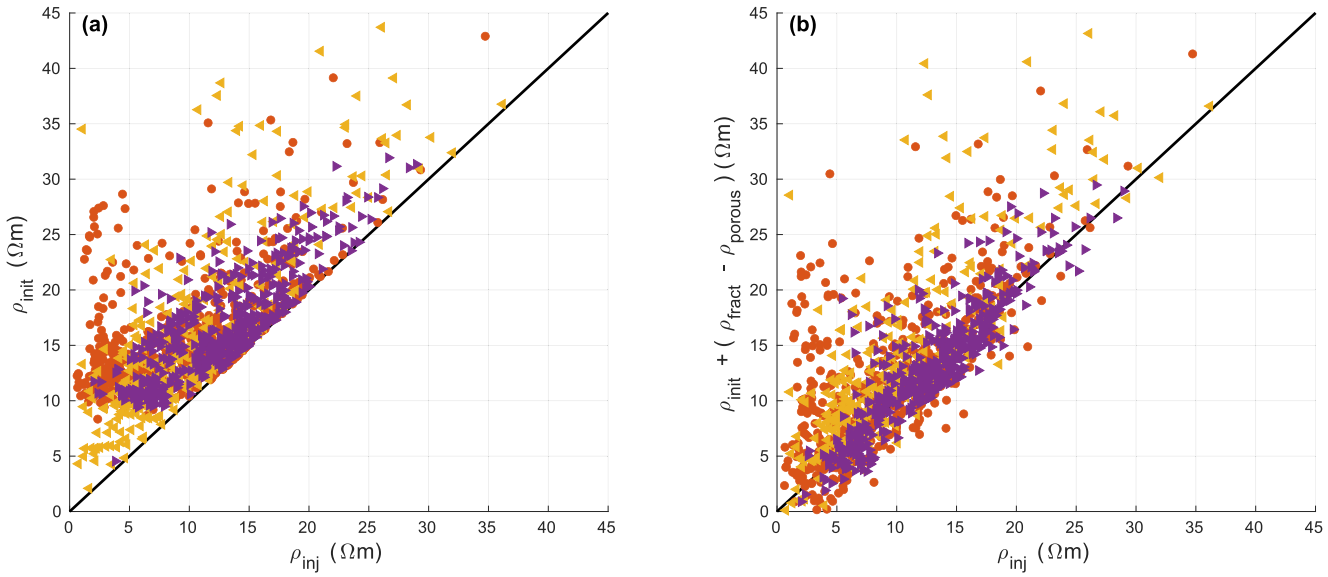


Figure 3. Scatter plot of the apparent resistivity observed after injection (ρ_{inj}) compared with (a) the apparent resistivity observed before injection (ρ_{init}), and (b) the term $\rho_{init} + (\rho_{fract} - \rho_{porous})$ representing inverted fractures added to the observed baseline (cf. Section 2.4). Quadrupole configurations are distinguished by various markers: single (orange circle), parallel (yellow left triangle), equatorial (purple right triangle).

This percentage also highlights some limitations of the model (cf. Table 1). Besides the iterative architecture of the inversion, other sources of differences may include the heterogeneous nature of the matrix. Our current discrete model assumes a homogeneous matrix but the variability in the baseline suggests non-uniform apparent resistivities (Figure 3a). Initial assumptions to limit the size of the parameter space (fixed fracture aperture, dip and conductivity, cf. Section 2.6), as well as the presence of low resistivity features outside the DFN (i.e., reagent spreading in the matrix), may hinder a further decrease in the misfit.

Finally, it cannot be ruled out that the electrode boreholes play themselves a role of preferential pathways for the injected product (Lévy, Thalund-Hansen, Bording, Fiandaca, Christiansen, et al., 2022). Vertical fractures close to the electrode boreholes may particularly affect single borehole data, which have the greatest sensitivity to resistivity variations close to the boreholes. The best fit is obtained with equatorial electrode configuration and the worst with the parallel (Figure 3b), in accordance with noise sensitivity in XB-ERT measurements (Zhou & Greenhalgh, 2001).

4. Conclusions

While standard inversion schemes used to invert electrical resistivity data are not well suited to study systems with thin localized flow-paths, we present a discrete inversion approach dedicated to this kind of heterogeneities. Despite the limitations induced by the iterative implementation of the inversion method and the a priori hypotheses, the presented approach succeeds in locating thin conductive structures at a contaminated site in Denmark undergoing remediation. Compared with standard inversion methods, the discrete inversion scheme does not rely on smoothing constraints and requires the inversion of fewer parameters. Future applications will benefit from a sensitivity analysis of the model parameters. Considering a change in dip and a heterogeneous matrix would be of great interest to improve the inversion strategy. Sensitivity to the number of datapoints should also be further evaluated, as the discrete and standard inversions differ in the ratio between parameters and data. For example, a data weighting could be considered depending on the measurement errors in the inversion of time-lapse data (Lesparre et al., 2017), or according to the datapoint locations relatively to the fractures.

Proof-of-concept should be improved by using a non-iterative implementation. The transdimensional method (e.g., Ringel et al., 2021; Römhild et al., 2024; Somogyvári et al., 2017) would be suitable as it presents the advantage of working with the number of fractures as a parameter to estimate. It will also fill the gaps in the uncertainty estimates of the current version of the model. Such stochastic methods need to deal with

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computational cost issues. The current inversion procedure takes around 150 hr over 112 threads, and would require further optimization.

Finally, long-term perspectives include the coupling between the considered discrete electrical model and flow and transport models in order to invert tracer experiments monitored with ERT data. Both borehole and surface electrical data could be considered with hydro-geochemical data such as pressure and tracer concentration monitored in boreholes under ambient and forced hydraulic conditions. The development and implementation of joint inversion will be needed with possible applications to fractured aquifers and karst systems.

Data Availability Statement

More information about the field site and conventional inversion of associated data is found in Lévy, Thalund-Hansen, Bording, Fiandaca, Christiansen, et al. (2022). DCIP data and inversion results with AarhusInv software, as well as protocol and spread files for the ABEM Terrameter LS2, are available in Lévy, Thalund-Hansen, Bording, & Fiandaca (2022). MontpInv software has been used to invert these data (Lelimouzin et al., 2024).

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